

CSC-335

HCI & Computer Graphics

Lec 4: UX Components and User Testing

Methods of UX Research



Three Foundational Methods of UX Research

ASK

Understand users' perspectives by asking them directly about their thoughts, needs, and behaviors.

OBSERVE

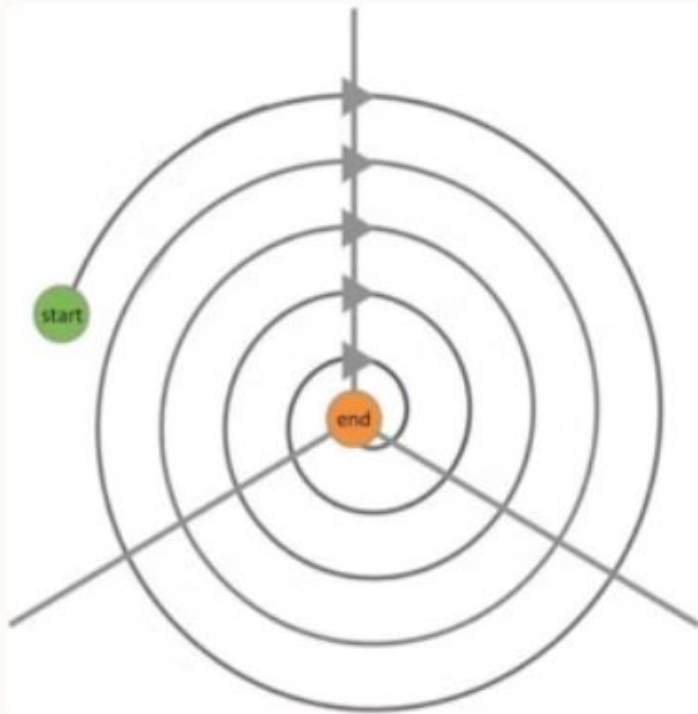
Gain insights by watching users interact with a system or perform tasks in their natural environment.

INSPECT

Evaluate designs and prototypes against established guidelines and best practices to identify potential usability issues.



Comprehensive UX Research Framework



Assess

Interviews, Surveys, Comparative Analysis

Design

Lo-Fi Testing, Hi-Fi Testing

Build

Heuristic Evaluation, Analytics, Observations

A multi-faceted approach combines various techniques across the product development lifecycle to ensure a holistic understanding of user needs and system performance.

The 3 Methods



ASK



OBSERVE



INSPECT



Ask: Gathering User Perspectives

Interviews

Structured conversations with stakeholders to deeply understand their experiences, motivations, and challenges.

Surveys

Distributed questionnaires to a large audience to collect quantitative and qualitative data on attitudes, behaviors, and demographics.

Focus Groups

Facilitated discussions with small groups of users to explore specific topics and gather collective opinions.

Diary Studies

Participants record their experiences, thoughts, and activities over an extended period to capture real-time usage and context.

Experience Sampling

Collecting data on users' experiences at random intervals in their natural environment, often via mobile devices.

Observe: Understanding User Behavior

Ethnographic Observations

Watching users in their natural environment to understand how they interact with products and perform tasks without intervention.

User Testing

Observing users as they complete specific, scripted tasks within a system to identify usability issues and assess task completion rates.

Usage Analytics

Analyzing large datasets of user interactions (e.g., clicks, navigation paths) to uncover patterns and trends in system usage.

Video Analysis

Reviewing recorded user sessions to meticulously analyze interactions, gestures, and expressions, providing rich qualitative data.

Social Media Mining

Extracting and analyzing publicly available data from social media to understand user sentiment, feedback, and trends related to a product.



Inspect: Evaluating Designs

Guideline-Based

Systematic evaluation of a design against established usability heuristics, accessibility standards, or industry best practices.

Walkthroughs

Simulating user interaction paths step-by-step to uncover potential friction points or breakdowns in the user journey.

Comparative Analysis

Assessing a product's design by comparing it against competitors or similar solutions to pinpoint strengths, weaknesses, and opportunities.

Hybrid Approaches: Watch and Ask

Often, combining observation with direct questioning yields richer insights, capturing both explicit and implicit user behaviors and motivations.

- **User Testing + Interviews** Observing users perform tasks, then asking follow-up questions to understand their decision-making and rationale.
- **Contextual Interviews** Interviewing users in their natural environment while they are engaging in relevant activities, capturing insights in real-time context.
- **Artifact-based Methods** Analyzing existing user-generated artifacts (e.g., notes, sketches, logs) to understand workflows and challenges, often combined with interviews about these artifacts.



When to Apply Which Method?

Ask when...

- Observation is infeasible due to infrequent, long-duration, or private activities.
- Understanding user values, motivations, and emotional responses is key.
- Large numbers and high certainty are needed (e.g., through surveys).

Observe when...

- Understanding actual behavior, not just reported behavior, is crucial.
- Identifying usability issues in context is important.
- Discovering unforeseen usage patterns is desired.

Inspect when...

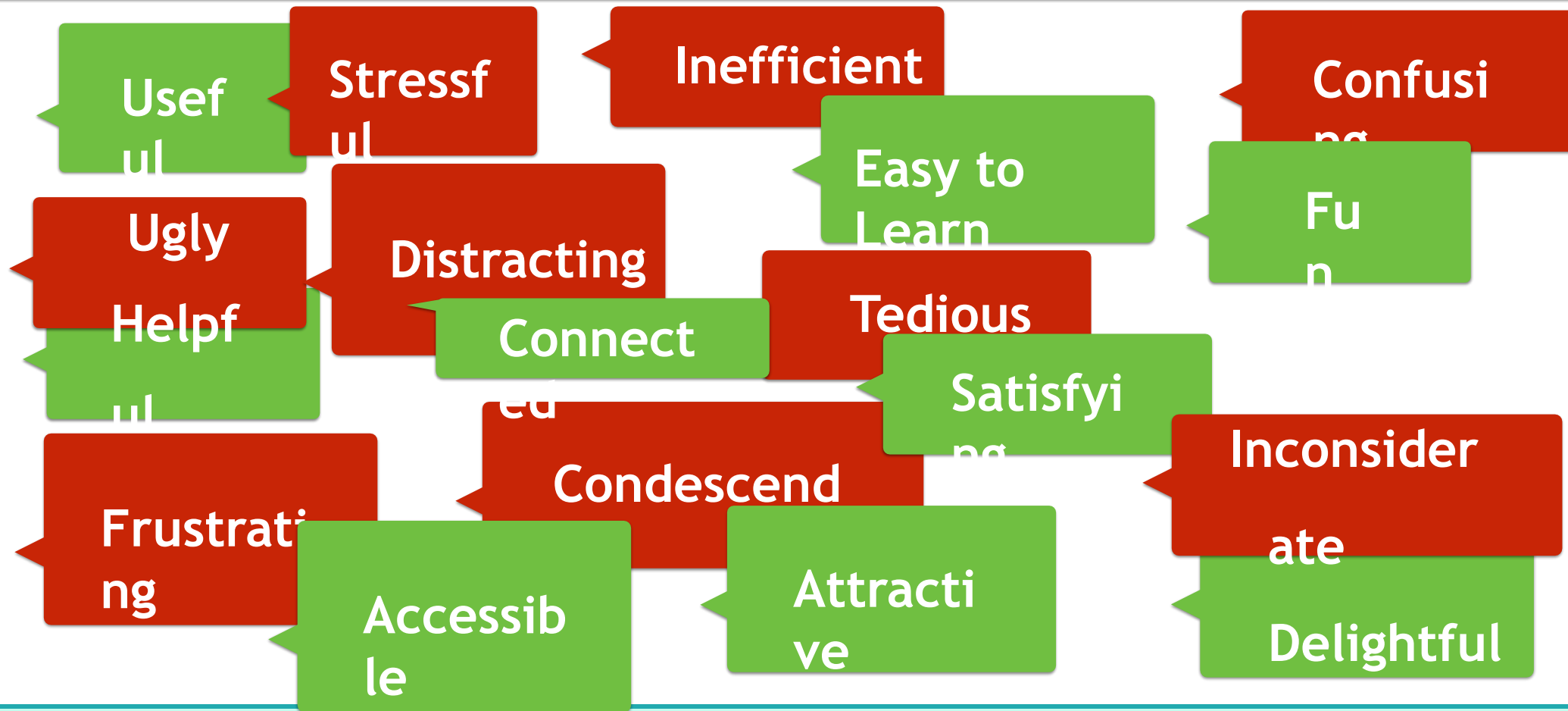
- There is a need for a quick and cost-effective expert evaluation.
- Benchmarking against established standards or competitors is required.
- Identifying probable flaws before extensive user testing is beneficial.



WHAT MAKES A GOOD USER EXPERIENCE?



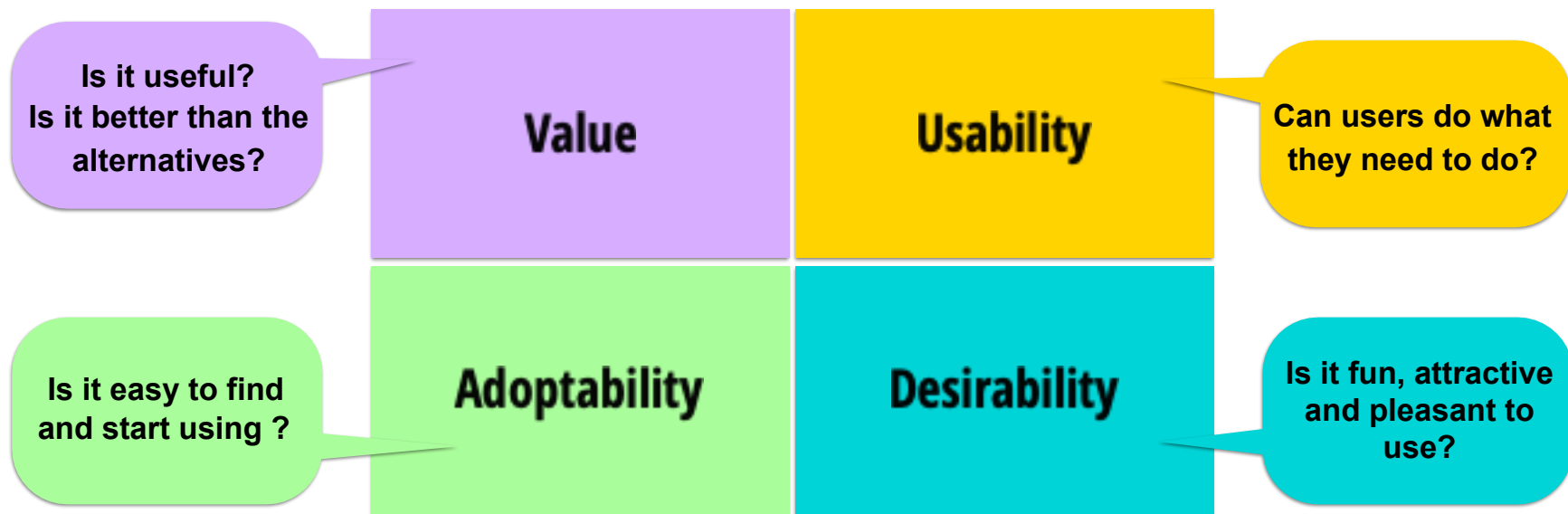
Good and Bad UX



Components of UX



Components of UX



VisiCalc 1979

C11 (L) TOTAL C125

	A	B	C	D
1	ITEM	NO.	UNIT	COST
2	---	---	---	---
3	MUCK RAKE	43	12.95	556.85
4	BUZZ CUT	15	6.75	101.25
5	TOE TONER	250	49.95	12487.50
6	EYE SNUFF	2	4.95	9.90
7				
8			SUBTOTAL	13155.50
9			9.75% TAX	1282.66
10			TOTAL	14438.16

MOTOROLA ROKR



iPod 2001

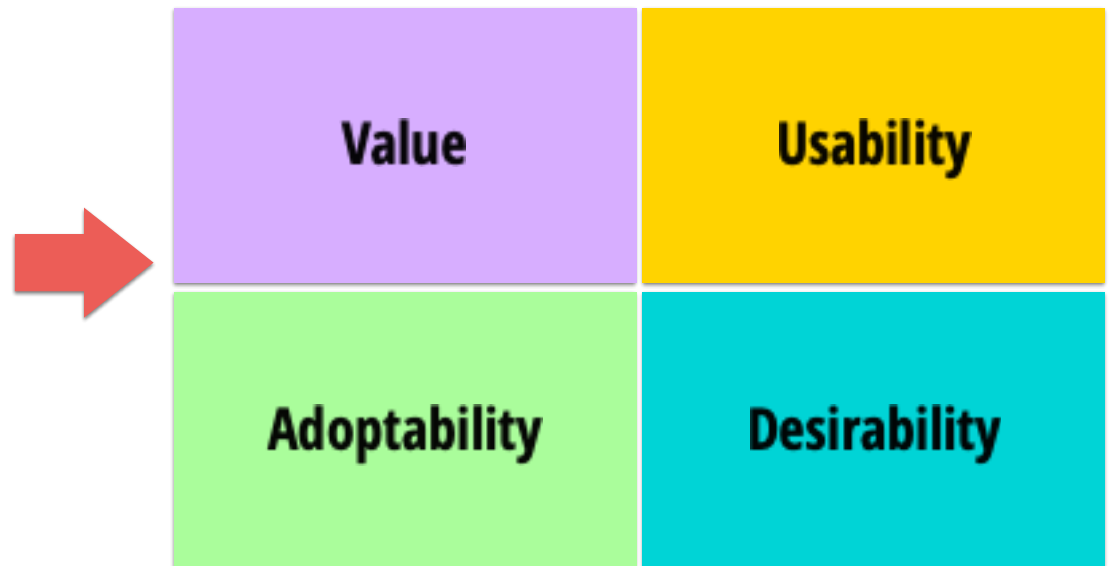


Adoptability - Duolingo Onboarding Process



Basic Methods of UX

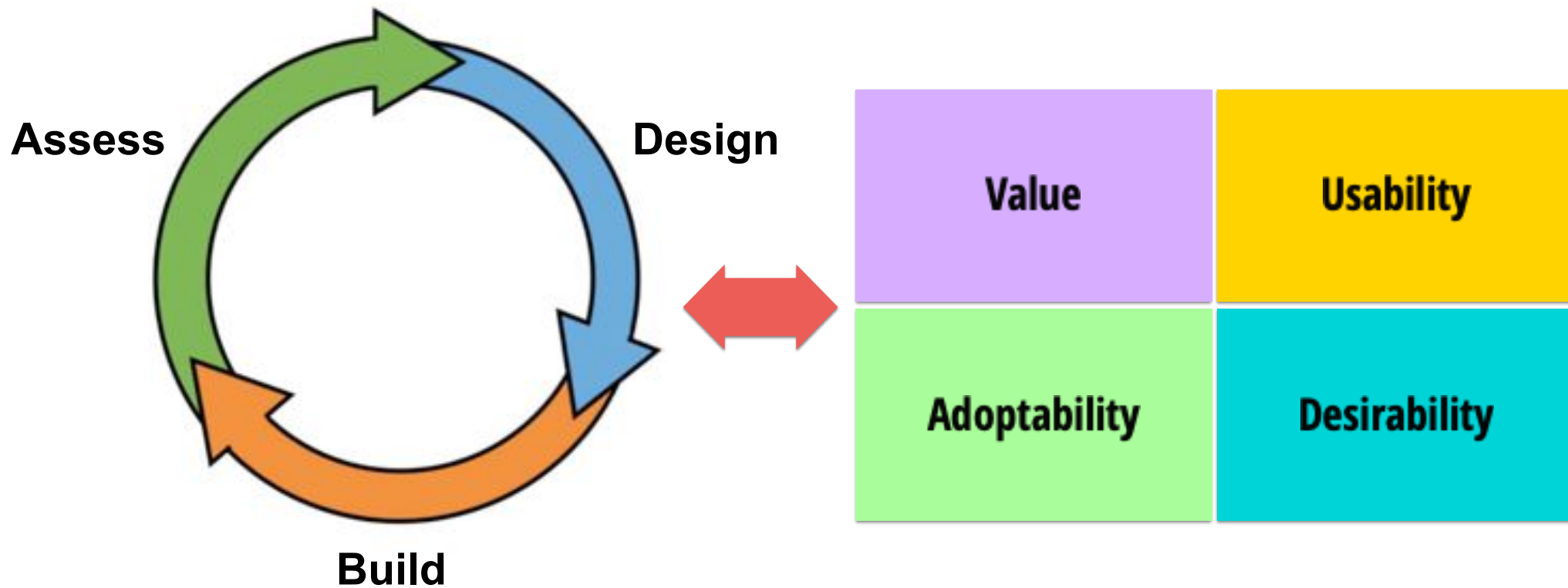
- Understanding Users
- Design & Prototyping
- Evaluating Designs



Assessing UX: Questions

	Understanding Users	Evaluating Designs
Value	What do users need?	Does this design fulfill the need?
Usability	How do they do it now?	Can they get it done with this?
Desirability	What do they desire?	Is the design appealing?
Adoptability	Where do users look for things?	Can users find and access this?

The Process is your Guide!





The 7 Factors that influence User Experience: <https://www.interaction-design.org/literature/article/the-7-factors-that-influence-user-experience>

The UX Honeycomb

Peter Morville



The UX Honey Comb: http://semanticstudios.com/user_experience_design/

The UX Honeycomb

